Project Demo Planning

Date	03 July 2026
Team ID	
Project Name	Predicting Human Development Index (HDI) Using Machine Learning
Maximum Marks	1 Mark

Project Demo Planning:

The project demonstration is planned to showcase the complete workflow of the Human Development Index (HDI) prediction system. The demonstration will explain the problem statement, data preprocessing, machine learning model development, Flask web application, prediction process, and system outcomes. The objective is to demonstrate how the application predicts HDI scores using socio-economic indicators and provides tier-specific policy recommendations.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction	Introduce the project objectives, motivation, and the importance of predicting HDI using Machine Learning. Explain how the system helps policymakers and international organizations predict development status in real-time.	1–2 mins	Swapna Beldhari
2	Solution Overview	Explain the overall system architecture, dataset, preprocessing steps, machine learning models evaluated (Linear Regression, Decision Tree, KNN), and Flask-	2 mins	Yaswanth Yadav Keerthi

		based web application.		
3	Live Application Demonstration	Launch the Flask application, enter sample development parameters, generate HDI prediction, and explain the gauge score, tier classification, and recommendations displayed by the system.	2 mins	Sai Kiran Bandi
4	Machine Learning Explanation	Briefly explain data preprocessing, label encoding, model comparison, evaluation metrics (R2, RMSE), and why the Linear Regression model was selected with R2 = 0.8834.	1–2 mins	Yaswanth Yadav Keerthi
5	Testing & Validation	Demonstrate functional testing, performance testing, input validation, prediction accuracy (e.g. clamping behavior), and application reliability.	2 mins	Sai Kiran Bandi
6	Future Enhancements	Discuss future improvements including cloud deployment, real-time World Bank/UNDP API integration, database integration for prediction history, and conclude the demonstration.	1 min	Swapna Beldhari

Total Duration - 8-10 mins

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain the need for an AI-assisted HDI prediction system, challenges of conventional HDI reporting, and project objectives.
2	Solution Overview	Present the system architecture, HDI dataset, technologies used, Machine Learning workflow, and Flask application architecture.
3	Live Feature Demonstration	Enter sample socio-economic parameters, generate HDI prediction, explain the circular gauge, tier classification, and policy recommendations.
4	Q&A Session	Answer questions related to Machine Learning algorithms, Flask implementation, dataset preprocessing, model evaluation, project design decisions, and future scope.